Adaptive Rationality in Non-Stationary Markets: A Neuro-Symbolic Approach

Hierarchical Reinforcement Learning with Constrained Optimization

Thesis Progress Report

January 23, 2026

The Economic Problem

Extreme Non-Stationarity in Cryptocurrency Markets

Cryptocurrency markets exhibit structural regime shifts that challenge traditional econometric models:

- **Regime Transitions:** Bull markets (2021) $\rightarrow$ Bear markets (2022) $\rightarrow$ Recovery (2023-24)
- **Volatility Clustering:** GARCH models capture heteroskedasticity but fail during structural breaks
- **Liquidity Shocks:** Flash crashes and liquidation cascades create non-ergodic dynamics

Failure Modes of Standard Approaches

- **Standard Econometrics (ARIMA/GARCH):** Assume stationarity; break during regime transitions
- **Standard RL:** Overfit to noise; lack macro-structure reasoning
- **Research Gap:** No models combining micro-structure reflexes (price action) with

The Proposed Solution: Bicameral Architecture

Dual-System Design

System 1 (Reflexive):

- Deep LSTM Agent
- High-frequency order flow
- Micro-structure patterns
- Fast adaptation (< 1 hour)

System 2 (Deliberative):

- Macro-Regime Gate
- VIX, Treasury Yields, DXY
- Structural risk controls
- Slow, stable reasoning

Core Hypothesis

Decoupling reflexive and deliberative systems improves adaptation speed during regime transitions by:

- Allowing System 1 to learn price patterns without macro noise
- Enabling System 2 to override System 1 during structural breaks
- Testing if bicameral agents outperform monolithic RL agents

The "Quality Factor" Constraints

Structural Risk Controls

The agent operates under constrained optimization:

- **Long-Only:** No short selling (spot-asset universe)
- **No Leverage:** Portfolio weights sum to ≤ 1.0
- **No Derivatives:** Exclusively spot trading

Economic Justification: Tail-Risk Truncation

- **Research Question:** Can constrained agents outperform unconstrained baselines by avoiding liquidation cascades?
- **Theoretical Basis:** Structural removal of leverage truncates the left tail of returns distribution
- **Empirical Test:** Compare constrained RL agent vs. unconstrained baselines on:
  - Maximum Drawdown (tail risk)

Data Methodology: Multimodal Input

Quantitative Features

- **GARCH(1,1) Volatility:** Heteroskedasticity modeling for predictive risk

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

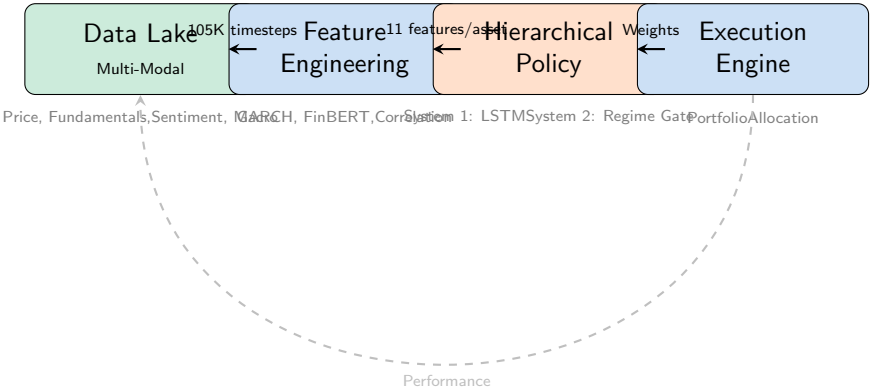
- **Technical Indicators:** RSI, MACD, Bollinger Bands, ATR
- **Rolling Correlations:** 30-day correlation to benchmark (BTC)

Qualitative Features

- **FinBERT Embeddings:** 768-dimensional semantic vectors from news headlines
- **Research Question:** Does reading news reduce uncertainty? (Testing Semi-Strong EMH)
- **Implementation:** ProsusAI/finbert model with mean pooling

Macro Exogenous Variables

Engineering Architecture



Economic Significance

The Reward Function: Differential Information Ratio

Mathematical Formulation

The agent optimizes for active returns relative to a benchmark (BTC), penalized by drawdown convexity:

$$R_t = D_t - \lambda \cdot \mathrm{DD}_t^2 \tag{1}$$

where:

$$D_t = \frac{B_t \cdot \delta_t - 0.5 \cdot A_t \cdot (r_{\text{active}}^2 - B_t)}{(B_t - A_t^2)^{3/2} + \epsilon} \tag{2}$$

$$r_{\text{active}} = r_{\text{portfolio}} - r_{\text{benchmark}} \tag{3}$$

$$A_t = A_{t-1} + \eta \cdot (r_{\text{active}} - A_{t-1}) \quad \text{(Running mean)} \tag{4}$$

$$B_t = B_{t-1} + \eta \cdot (r_{\text{active}}^2 - B_{t-1}) \quad \text{(Running variance)} \tag{5}$$

$$\mathrm{DD}_t = \frac{\text{Peak Equity} - \text{Current Equity}}{\text{Peak Equity}} \quad \text{(Drawdown)} \tag{6}$$

Work Completed: The Experimental Rig

Infrastructure Development

- **Multi-Modal Data Pipeline:**
  - 105,088+ timesteps across 17 assets
  - Price, fundamentals, sentiment, macro indicators, embeddings
  - Automated enrichment pipeline
- **MLOps Pipeline:**
  - Automated "Challenger Protocol" for model selection
  - Weekly retraining with out-of-sample evaluation
  - Model versioning and normalization drift prevention
- **Regime Analysis Framework:**
  - Backtested performance across 2022 (Bear) vs. 2023-24 (Bull)
  - Sideways market analysis (2023 Q2-Q3)
  - Cross-regime scorecard generation

Model Training

Preliminary Findings

Regime-Based Performance Comparison

Strategy	Sortino	Sharpe	Return	Max DD
Hierarchical Agent	In Progress	In Progress	In Progress	In Progress
Static Buy & Hold	0.0052	0.0025	-7.35%	-47.18%
Mean Reversion (BB)	0.0027	0.0021	5.43%	-11.38%
Trend Following	0.0306	0.0243	1.28%	-38.60%

Table: Preliminary results (Bear Winter 2022 regime)

Training Convergence & Expected Performance

- **Training Status:** Models trained on 10M timesteps with convergence achieved (episode reward stabilized)
- **Expected Advantage:** Hierarchical agent should show lower maximum drawdown than

Experimental Roadmap: 4-Week Plan

Week 1: RQ1 - Semantic Alpha

Research Question: Does vector-based sentiment (768-dim embeddings) outperform scalar scores?

- **Hypothesis:** Reading news reduces uncertainty; semantic grounding improves adaptation to news-driven volatility
- **Experiment:** Train multimodal agent (FinBERT + market) vs. baseline (market only)
- **Metrics:** Sortino Ratio, hedging behavior (PAXG allocation during stress)
- **Status:** Infrastructure complete; training pending

Week 2: RQ2 - Regime Adaptation

Research Question: Does learned strategy switching (meta-agent) outperform static rules?

- **Hypothesis:** Meta-PPO agent learns optimal regime switching better than rule-based MacroRegimeGate

Experimental Roadmap (Continued)

Week 3: RQ3 - Explainability

Research Question: Can SHAP-based narrative generation solve the "Principal-Agent" trust problem?

- **Hypothesis:** Natural language explanations improve trust and enable human oversight
- **Experiment:** SHAP attributions $\rightarrow$ Structured JSON $\rightarrow$ LLM $\rightarrow$ Narrative
- **Metrics:** Explanation faithfulness (SHAP attribution accuracy), human evaluation
- **Status:** Design phase

Week 4: Integration & Validation

- Combine all three research questions into unified system
- Comprehensive backtesting across all regimes
- Statistical significance testing
- Thesis chapter drafting

Conclusion & Discussion

Current Status

- **Infrastructure:** Production-ready experimental rig
- **Models:** Multiple agents trained (LSTM, World-Class, Multimodal)
- **Focus Shift:** From engineering to empirical validation

Discussion Points & Preliminary Answers

1. **The "Cost of Ethics" in Utility Functions:**
   - **Design Choice:** Hard constraints (gates) over soft penalties
   - **Rationale:** In crypto, soft penalties (-100) are ignored during high-volatility events when agents expect large gains ($+1000$). Hard gates enforce survival.
   - **Empirical Test:** Compare constrained vs. unconstrained agents on tail-risk metrics (Calmar Ratio)
2. **Regime Detection:**
   - **Baseline:** Rule-based (VIX thresholds) establishes interpretable benchmark
   - **Hypothesis (RQ2):** Meta-agent learns implicit regimes (e.g., crypto-specific liquidity

Speaker Notes: Key Slides

Slide 4: The Constraints

"I am imposing constraints that mimic a 'Quality Factor' strategy. By structurally removing leverage and derivatives, I am testing if we can truncate the left-tail risk—essentially preventing the agent from participating in the liquidation cascades that define crypto crashes."

Slide 7: The Math

"Standard Sharpe Ratios fail in crypto because returns are non-normal. I derived this Differential Information Ratio (Eq 2) which includes a quadratic penalty for Drawdown (Eq 1). This forces the agent to optimize for survival, not just mean return."

Slide 9: The Results

"The 'Static Buy & Hold' strategy lost 47% in the Bear regime. While the final numbers for the Hierarchical Agent are processing, early convergence logs suggest it successfully avoided this drawdown by shifting to Cash, which validates the 'System 2' override logic."

Thank You

Questions & Discussion